

Amphiphilic Marine Antifouling Coatings Based on a Hydrophilic Polyvinylpyrrolidone and Hydrophobic Fluorine–Silicon-Containing Block Copolymer

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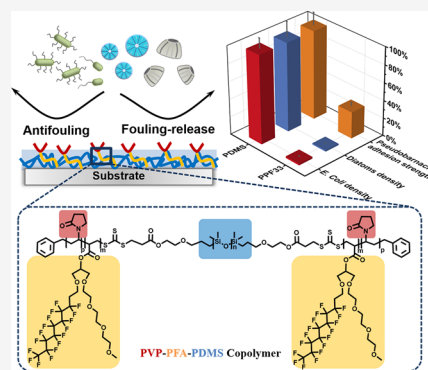


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Supporting Information

ABSTRACT: The development of environmentally friendly and highly efficient antifouling coatings is vastly desirable in the marine industry. Herein, we prepared a novel amphiphilic block copolymer that combined hydrophilic polyvinylpyrrolidone (PVP) with hydrophobic poly(1-(1*H*,1*H*,2*H*,2*H*-perfluorodecyloxy)-3-(3,6,9-trioxadecyloxy)-propan-2-yl acrylate) (PFA) and polydimethylsiloxane (PDMS). The amphiphilic copolymer (PVP–PFA–PDMS) was blended into a cross-linked PDMS matrix to form a set of controlled surface composition and surface-renewal coatings with efficient antifouling and fouling-release properties. These coatings incorporated the biofouling settlement resistance ability attributed to the hydrophilic PVP segments and the reduced adhesion strength attributed to the low surface energy of fluorine–silicon-containing segments. As expected, the coatings showed an excellent antifouling performance against bacteria and marine unicellular *Navicula parva* diatoms (98.1 and 98.5% of reduction, respectively) and fouling-release performance against pseudobarnacle adhesion (84.3% of reduction) compared to the pristine PDMS coating. Moreover, a higher-content PVP-based coatings presented higher ability to resist biofouling adhesion. The nontoxic antifouling coating developed in this paper hold the potential to be applied in a variety of marine industrial facilities.



INTRODUCTION

Marine biofouling is formed by the settlement and accumulation of microorganisms on surfaces of equipment immersed in seawater.^{1,2} It is a worldwide problem leading to huge economic loss and negative environmental effects on maritime industries.^{3,4} It can increase the roughness of the ship surface and generate an increasing frictional drag, consequently leading to power penalties, increased fuel consumption, and greenhouse gas emissions.^{1,5,6} For instance, it is reported that the use of efficient antifouling protection can save \$150 billion for the marine industry each year.² Currently, antifouling paints containing biocidal compounds are effective to prevent marine biofouling.^{7,8} However, biocidal compounds commonly exhibit negative effects on nontarget organisms in marine environments.^{9–11} To the environmental concerns considered, biocide-based coatings become more regulated, resulting in an increased demand for environmentally friendly and high-performance alternatives.^{8,12}

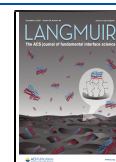
Ideal marine coatings should be able to simultaneously resist the initial settlement of marine organisms (antifouling property) and have low adhesion strength with the organisms (fouling-release property) without environmental threats.^{2,13,14} Recently, many nontoxic strategies have been extensively explored to meet the challenge of marine biofouling.^{15–20} For example, the controlled surface topography is applied to

influence the settlement behavior of marine organisms;^{21–25} the surfaces are modified by fouling-deterrent bioactive molecules, or antifouling zwitterionic groups, and so forth.^{26–31} Modifications of zwitterionic materials have achieved excellent mechanical stability, tunable elasticity, or long-term anti-biofilm properties.^{32–34} In recent years, amphiphilic materials presenting both antifouling and fouling-release properties have been a promising marine coating.^{4,35–37} The amphiphilic coating surfaces incorporate the hydrophilic segments (e.g., polyethylene glycol, PEG) to resist organism attachment with the hydrophobic segments (e.g., polydimethylsiloxane, PDMS) to reduce the adhesion strength.^{38–41} However, the ether linkage in PEG is susceptible to thermal/oxidative chain scission, significantly reducing the protein resistance.^{42–45} Moreover, hydrophobic PDMS cannot prevent the formation of marine slime layers (mainly composed of protein, bacteria, and diatoms) because of the

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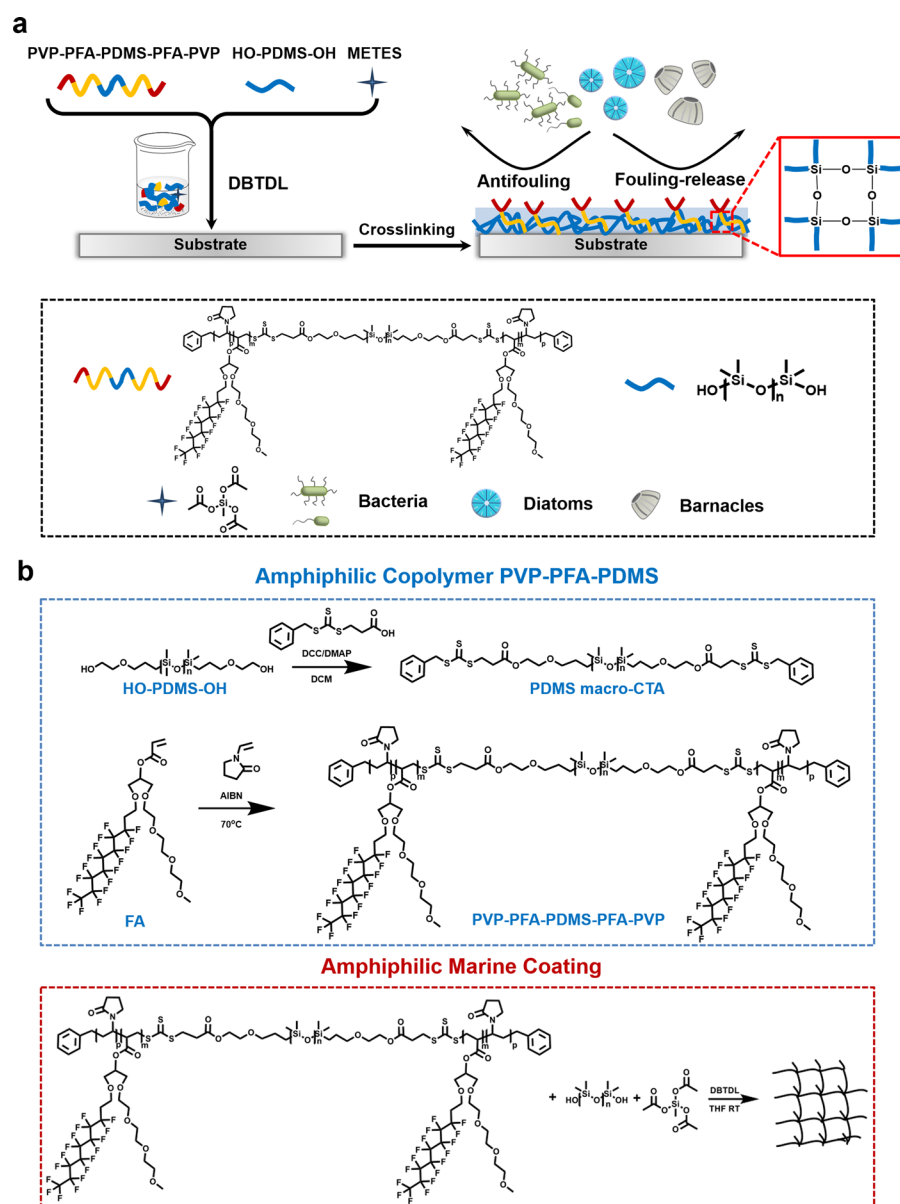


Figure 1. (a) Schematic illustration for the design of amphiphilic marine antifouling coating based on the PVP–PFA–PDMS block copolymer. (b) Synthesis of PDMS macromolecular chain transfer for block copolymerization by RAFT with NVP and FA, and preparation of marine coating.

hydrophobic interaction between slime-based marine organisms and PDMS materials.^{46,47}

In this work, a new type of amphiphilic copolymer (PVP–PFA–PDMS) composed of polyvinylpyrrolidone (PVP), poly(1-(1*H*,1*H*,2*H*,2*H*-perfluorodecyloxy)-3-(3,6,9-trioxadecyloxy)-propan-2-yl acrylate) (PFA), and PDMS is designed and synthesized. Subsequently, the amphiphilic copolymer is blended into a crosslinked PDMS matrix to develop a highly efficient marine antifouling coating. Because the pyrrolidone units can tightly bind water molecules, PVP presents a high hydrophilicity to form a strong hydration layer, which can eliminate the electrostatic force and hydrophobic interaction between the coating surface and biomolecules.^{35,48} Therefore, hydrophilic PVP segments in PVP–PFA–PDMS copolymer can effectively resist the settlement of fouling organisms. More importantly, PVP has been reported to possess a superior long-term stability even in harsh conditions (e.g., 3% of H₂O₂ solution for 100 h at 60 °C).⁴⁹ Meanwhile, the incorporated

PFA (a fluorine–silicon-containing segment) serves as a hydrophobic unit in the copolymer, aiming to further reduce the surface energy and resist slime adhesion, thereby promoting release of fouling organisms.^{50,51} The PDMS segment in the copolymer can ensure the compatibility with a PDMS crosslinking matrix, forming a novel amphiphilic marine antifouling coating.^{52,53} The resultant coatings presented excellent antifouling and fouling-release properties against the adhesion of marine organisms, including bacteria, unicellular *Navicula parva* diatom, and barnacle, as shown in Figure 1a.

EXPERIMENTAL SECTION

Materials. Hydroxyl-terminated PDMS (5.6k HO-PDMS-OH and 26k HO-PDMS-OH) were purchased from Gelest. Dicyclohexylcarbodiimide, *tert*-butyl(chloro)dimethylsilane, 4-(dimethylamino)pyridine, triethoxymethyl-silane (purified: 96%), hexane, ethyl acetate, and methanol were all obtained from Heowns. *N*-Vinylpyrrolidone (NVP, Acros, 99% stabilized with NaOH) and dichloromethane

Table 1. Characterization of the Amphiphilic Block Copolymers

polymer	NVP/FA/PDMS-CTA ^a	NVP/FA/PDMS-CTA (NMR) ^b	M _n (NMR) ^c (g/mol)	M _n (target) ^d (g/mol)
PVP-PFA-PDMS-1	212/6/1	62/2/1	12,644	30,896
PVP-PFA-PDMS-2	258/8/1	144/4/1	24,368	40,316
PVP-PFA-PDMS-3	337/14/1	190/6/1	31,088	53,750

^aComonomer NVP/macroinitiator mole ratio in the initial feed. ^bMolecular ratio of repeat units in the copolymer by nuclear magnetic resonance (NMR). ^cMolecular weight calculated from NMR. ^dMolecular weight calculated from target (in kg mol⁻¹).

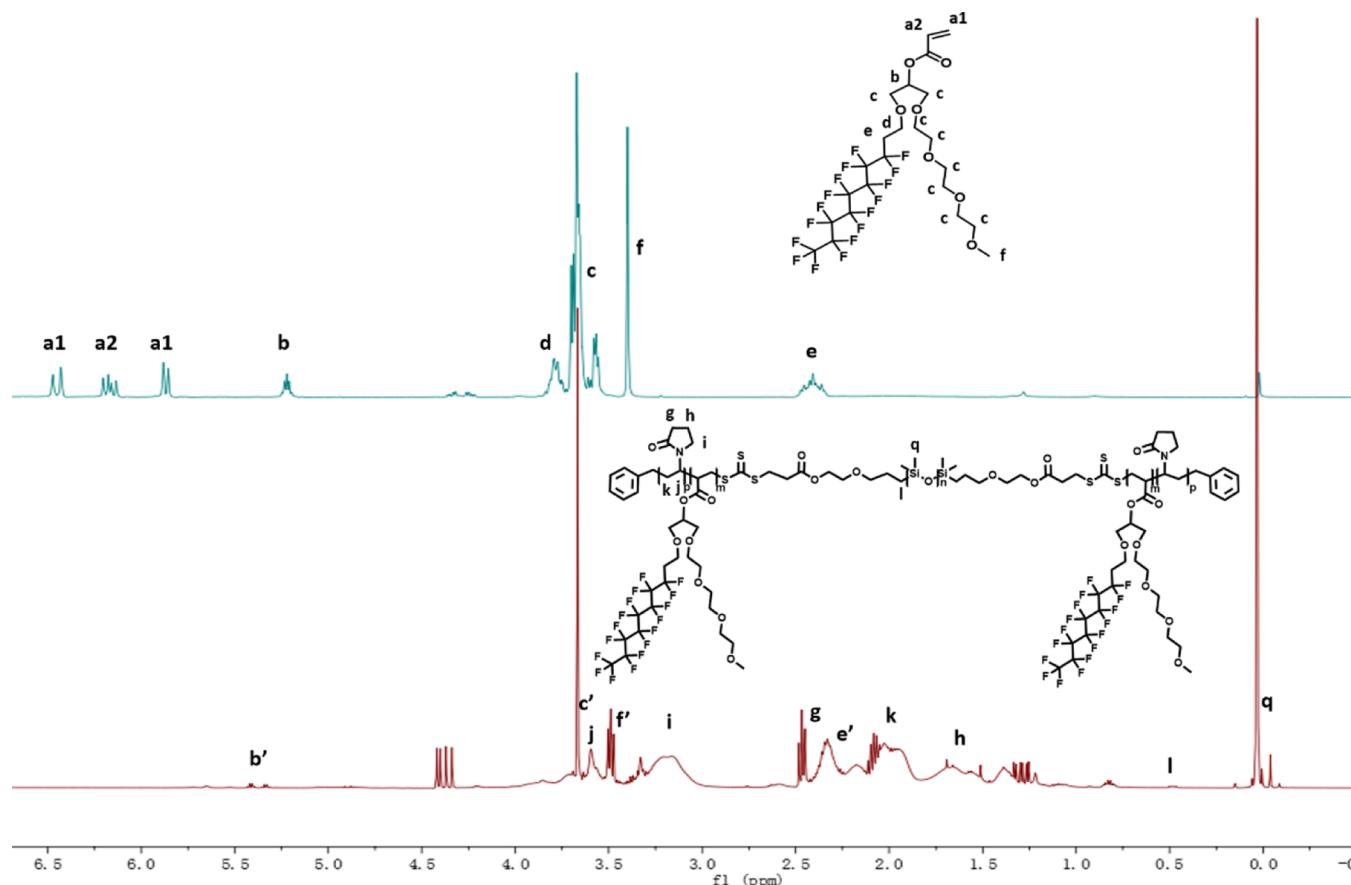


Figure 2. ¹H NMR spectra of FA and PVP-PFA-PDMS. ¹H NMR (CDCl₃, δ ppm): 0.01–0.16 (m, 6H, Si(CH₃)₂-O); 0.48–0.53 (t, 2H, Si(CH₃)₂-CH₂); 1.21–1.80 (m, 2H, N-CH₂-CH₂-CH₂-CO); 1.80–2.13 (m, 2H, CH₂-CHNC); 2.13–2.51 (m, 2H + 2H, N-CH₂-CH₂-CH₂-CO and O-CH₂-CH₂-CF₂); 3.05–3.48 (m, 2H + 3H, N-CH₂-CH₂-CH₂-CO and O-CH₂); 3.48–4.00 (m, 2H + 16H + 2H, CH₂-CHNC, CH-(CH₂)₂-O-CH₂-CH₂-O-CH₂-CH₂-O-CH₂-CH₂-O and O-CH₂-CH₂-CF₂); 5.30–5.40 (m, O-CH-(CH₂)₂); 5.80–5.90 (CH₂=CH-CO); 6.10–6.23 (CH₂=CH-CO); 6.36–6.48 (CH₂=CH-CO).

(Sigma-Aldrich) were treated by vacuum distillation prior to the experiment. Dibutyltin dilaurate (DBTDL, 97.5%) and diiodomethane (CH₂I₂) were both purchased from J&K, China. 2, 2'-Azobis(isobutyronitrile) (AIBN) was obtained from Sigma-Aldrich. It was recrystallized in methanol for three times to remove impurities. 1-(1H,1H,2H,2H-perfluorodecyloxy)-3-(3,6,9-trioxadecyloxy)propan-2-yl acrylate (FA) was synthesized according to the literature reported.⁵⁴ Glass slides (Sail Brand, 20 × 50 mm²) were cleaned in acetone and dried. Artificial seawater (ASW, pH 8.1–8.3) was prepared according to American Society for Testing and Materials (ASTM) standard D 1141-98.⁵⁵ The density of the ASW is 1.024 g/cm³, and the concentration is 3.23%.

Synthesis of Amphiphilic PVP-PFA-PDMS Copolymers. Three types of amphiphilic block copolymers (PVP-PFA-PDMS) with different ratios (as shown in Table 1) were synthesized via reversible addition-fragmentation chain transfer (RAFT) copolymerization utilizing PDMS macromolecular chain transfer agent (macro-CTA). Difunctional macro-RAFT agents were prepared according to a previous report.⁵⁶ For example, to prepare copolymer P1, a solution composed of AIBN (8.2 mg, 0.5 mmol), PDMS macro-

CTA (5.62 g, 0.95 mmol), FA (4.21 g, 5.7 mmol), and NVP (22.22 g, 200.00 mmol) in 12 mL of 1,4-dioxane was prepared in a dried flask with magnetic stirring. In order to deoxidize the mixed solution, nitrogen was then bubbled at room temperature (RT) for 30 min. The reaction flask was immersed in an oil bath which was preheated to 70 °C for 24 h. Subsequently, it was quenched with an ice bath. The solution was concentrated, followed by precipitation in tenfold cold diethyl ether at least three times. After filtration, the solid was collected and dried in a vacuum drying chamber at 40 °C overnight.

Preparation of Amphiphilic Coatings. The two-step procedure of preparing amphiphilic coatings with PVP-PFA-PDMS block copolymer is summarized as follows. For example, to prepare coating PPF21, 5.0 g of the PDMS matrix, 0.125 g of crosslinker triethoxymethyl-silane, and catalyst DBTDL (50 mg) were dissolved in tetrahydrofuran (24 mL) and then the solution was deposited on glass slides. At RT, the coatings were desiccated on the first day and toughened at 120 °C on the second day. A solution of the same composition with PVP-PFA-PDMS-2 (200 mg) was coated on top of the first layer and cured as the above protocol. The ratios of amphiphilic block copolymers to the PDMS matrix were 4, 8, and 12

wt % in these coatings. The control group was a blank PDMS coating without block copolymers. The thickness of the coatings is ~ 0.5 mm.

Characterization. To confirm the successful synthesis of the copolymers, nuclear magnetic resonance (NMR) measurements (Bruker DRX-600 spectrometer, 400 MHz) were performed using CDCl_3 solution. Fourier transform infrared (FTIR) spectroscopy was carried out on an ALPHA II spectrometer (Bruker, Germany) at RT with a scan range of $400\text{--}4000\text{ cm}^{-1}$. PVP-PFA-PDMS copolymer/KBr powder tableting method was used for the FTIR measurement. In all cases, 24 scans at a resolution of 2 cm^{-1} were performed to obtain the infrared spectra. Water contact angles were recorded using the sessile-drop method. Both deionized water and CH_2I_2 were tested by an optical contact angle goniometry meter (Shanghai Zhongchen Digital Technology Apparatus Co. JC2000D1) at RT. For water and CH_2I_2 , the droplet volumes were 5 and $2.5\text{ }\mu\text{L}$, respectively. After five individual measurements on different surface areas of the same coating, the average value was taken as the reported contact angles. The Owens-Wendt method was used to calculate the surface free energy (γ_s) and its dispersive (γ_s^D) and polar (γ_s^P) components of the coatings.^{57–59}

Bacterial Adhesion Tests. The coating samples were sterilized and prepared as 0.7 cm diameter discs and placed in a 48-well cell-culture plate. Then, $100\text{ }\mu\text{L}$, 10^6 CFU mL^{-1} of the bacterial suspension was added into each well. The samples were gently rinsed for three times with 1 mL of PBS after a 4 h incubation at $37\text{ }^\circ\text{C}$. Using an ultrasound cleaner, $100\text{ }\mu\text{L}$ of the final washing solution was dispersed in the beef-protein medium and cultured in an incubator at $37\text{ }^\circ\text{C}$ for 24 h.⁶⁰ The bacteria removal percentage was calculated based on the counts of the PDMS and PPF samples with 95% confidence limits. Error bars represented the standard error of the mean.

Marine Diatom Attachment Assays. Before the measurement, three glass slides were equilibrated in filtered ASW for 72 h. A diatom *N. parva* suspension with $1 \times 10^6\text{ cells mL}^{-1}$ concentration was prepared.⁵¹ 10 mL of the diatom suspension was introduced on the coating surfaces in triplicate. After two hours, the samples were immersed in ASW to remove the unattached cells, and the diatoms on the surfaces were immediately observed under an optical microscope (Nikon Eclipse Ti-S). In the three sets of replicate samples, diatoms in 30 fields of view in each were counted, thereby providing statistical average data of the attached diatoms (95% confidence interval).

Pseudobarnacle Adhesion. The pseudobarnacle adhesion tests were performed as reported previously according to ASTM D 5618-20.⁶¹ Detailed procedure can be found in the Supporting Information.

RESULTS AND DISCUSSION

Preparation and Characterization of the Antifouling Coatings. As shown in Figures 1 and S1, the amphiphilic block copolymer combined the hydrophilic PVP with hydrophobic fluorine-silicon-containing segments. PDMS macro-CTA was used as the macro-chain transfer agent, and the copolymers were synthesized by RAFT polymerization of NVP and FA. Here, a hydrophilic PVP unit was proposed to improve the antifouling property of the resultant coatings, and an FA unit was proposed to further reduce the surface energy of coatings. Table 1 presents three molar ratios of copolymers, which were identified by abbreviated names PVP-PFA-PDMS-1 (PPF1), PVP-PFA-PDMS-2 (PPF2), or PVP-PFA-PDMS-3 (PPF3).

First, the copolymers were characterized by ^1H NMR and ^{19}F NMR spectroscopy (Figures 2 and S2). In H-spectrum, with respect to PVP-PFA-PDMS, the signals at 0.1 ppm represented the methyl protons of PDMS. $\delta\text{ }1.21\text{--}1.80$ represented the proton absorption peak of the 4-position methylene group of the pyrrolidone ring ($-\text{CH}_2-$). $\delta\text{ }2.13\text{--}2.51$ was attributed to the absorption peak of the methylene proton near the pyrrolidone ring $-\text{CO}-$ bond and the

methylene proton peak adjacent to $-\text{CF}_2-$ on the PFA segment. $\delta\text{ }3.05\text{--}3.48$ was contributed by the absorption peak of the methylene proton which was adjacent to the N atom on the pyrrolidone ring $-\text{NCH}_2-$ and the methyl proton peak on the PFA segment of the polymer. $\delta\text{ }3.48\text{--}4.00$ represented the proton peak of the methine group connected to N on the pyrrole unit and the methylene proton peak on the PFA segment. The proton absorption peak about the PFA mainly appeared in $\delta\text{ }2.13\text{--}2.51$, $3.48\text{--}4.00$, and $5.30\text{--}5.40$. The molecular weight (M_n) and relative compositions of the copolymer were determined by the peaks at $3.05\text{--}3.48\text{ ppm}$ (PVP segment), $3.48\text{--}4.00\text{ ppm}$ (PFA), and the absorption peak at 0.1 ppm (PDMS). The corresponding F-spectrum also demonstrated that FA successfully participated in the reaction (Figure S2).

Next, FTIR spectrometry was used to further evaluate the copolymer. Figure 3 shows that in all samples, the asymmetric

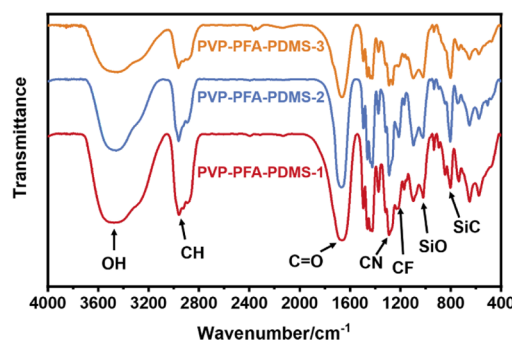


Figure 3. FTIR spectra of copolymer PVP-PFA-PDMS.

stretching vibrations of Si-O induced the absorption peak in the range of $1010\text{--}1120\text{ cm}^{-1}$. The peak observed at 802 cm^{-1} was assigned to the Si-C bending vibration, indicating the presence of PDMS. PFA was characterized by C-F stretching vibration bands at 1219 cm^{-1} . Due to the strong hydrophilicity of PVP segments, a water absorption peak in the range of $3360\text{--}3570\text{ cm}^{-1}$ could be observed. In addition, the peaks at 2960 , 1662 , and 1289 cm^{-1} presented the stretching-vibration of $-\text{CH}-$, $-\text{C}=\text{O}$, and $-\text{CN}-$, respectively. These results could demonstrate the successful polymerization of PVP-PFA-PDMS.

Static Contact Angle and Surface Free Energy. The topographical structures of the material surface and chemical compositions are the two critical factors to influence the wettability of the surface. In the first layer of coatings, the composition includes PDMS, triethoxymethyl-silane (2.5 wt % as a fraction of PDMS), and DBTDL (1 wt %). In the second layer, the composition includes PDMS, PVP-PFA-PDMS (4, 8 or 12 wt %), triethoxymethyl-silane (2.5 wt %), and DBTDL (1 wt %). Static contact angles of water and CH_2I_2 , besides surface free energies of different coatings are shown in Table 2. In addition, in order to study the time-dependent surface wettability, water contact angles of the coatings were measured after they were immersed in ASW for different periods ($t = 0, 24, 48, 72, 96$, and 120 h).

As shown in Figure 4, after continuously immersing in ASW, contact angles of all samples decreased and most of them stayed unchanged after 24 h. The addition of amphiphilic block copolymers could increase the hydrophilicity of the coating surfaces, so the initial contact angles ($t = 0$) of copolymer-based coatings were all lower than that of pristine

Table 2. Static Contact Angles and Surface Free Energies of Amphiphilic Coatings

	coating	polymer (wt % PDMS)	static contact angle (deg)		surface free energy (mJ m^{-2})		
			$\theta_{\text{H}_2\text{O}}$	$\theta_{\text{CH}_2\text{I}_2}$	γ_s	γ_s^D	γ_s^P
polymer 1	PDMS	0	109.5 ± 1.3	70.3 ± 1.5	23.0	22.9	0.1
	PPF11	4	100.3 ± 1.5	64.7 ± 1.5	25.9	25.1	0.8
	PPF12	8	95.8 ± 1.7	61.0 ± 1.0	28.2	26.7	1.5
	PPF13	12	91.2 ± 2.2	57.3 ± 0.6	30.6	28.3	2.3
polymer 2	PPF21	4	98.3 ± 1.0	64.7 ± 0.6	25.9	24.7	1.2
	PPF22	8	95.3 ± 1.0	63.7 ± 0.6	26.8	24.9	1.9
	PPF23	12	93.0 ± 2.2	61.3 ± 0.6	28.2	26.0	2.2
polymer 3	PPF31	4	99.0 ± 0.8	66.3 ± 0.6	25.0	23.8	1.2
	PPF32	8	96.3 ± 1.0	61.3 ± 2.1	28.0	26.6	1.4
	PPF33	12	92.5 ± 1.3	60.7 ± 2.1	28.6	26.3	2.3

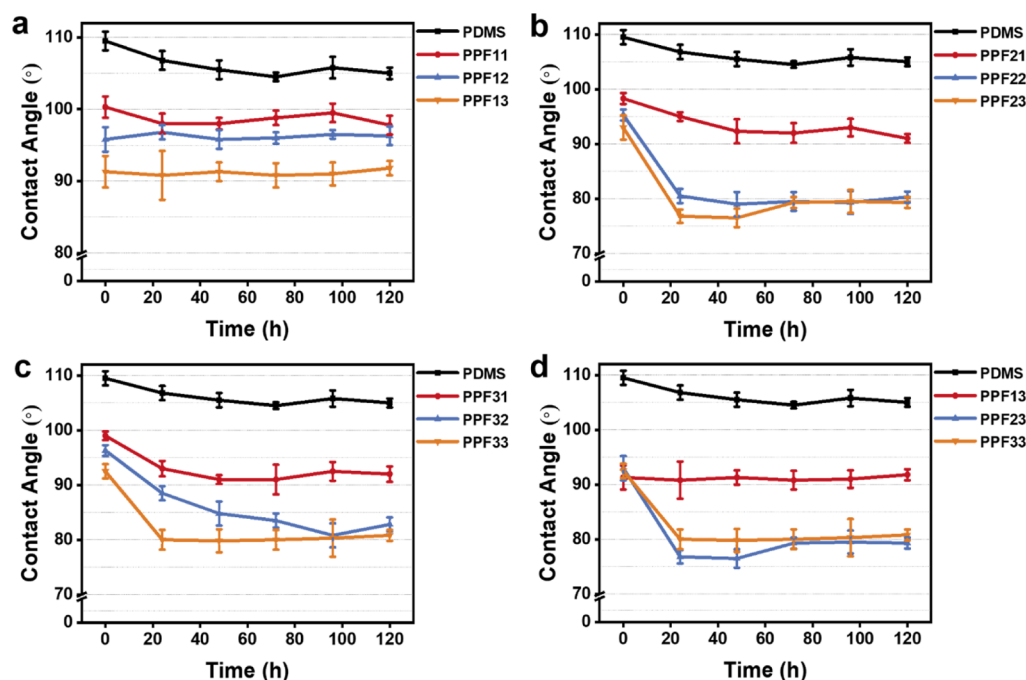


Figure 4. Contact angle measurements of (a) PPF1, (b) PPF2, (c) PPF3, and (d) PPF13–PPF33 coatings immersed in ASW.

PDMS coating. With the increasing time, the coatings containing the high content of amphiphilic copolymers with a higher molecular weight presented a significant decrease of contact angles, suggesting a surface-active property of the amphiphilic copolymer-based coatings. The hydrophilic PVP segments migrated from inside to the surface of the amphiphilic coatings, leading to their increased hydrophilicity. After 120 h immersion, contact angles of these coatings were inversely proportional to the contents of the hydrophilic PVP segments (Table 3).

Table 3. Contact Angle Measurements of Coatings after Immersion in ASW

sample	contact angle	sample	contact angle
PDMS	105.0 ± 0.8	PPF22	81.3 ± 1.0
PPF11	97.6 ± 1.3	PPF23	79.3 ± 1.0
PPF12	96.3 ± 1.3	PPF31	92.0 ± 1.4
PPF13	91.8 ± 1.0	PPF32	82.8 ± 1.3
PPF21	91.0 ± 0.8	PPF33	80.8 ± 1.0

In accordance with the Owens–Wendt method, the surface free energy of the coatings was determined by the contact angles of water and CH_2I_2 . The static θ_s and surface free energy γ_s of the PDMS and PPF coatings are presented in Table 2. It was shown that the surface free energy of PPF was higher than pristine PDMS coating because of the enhanced surface hydrophilicity by incorporation of PVP units. When the weight percentage of PVP–PFA–PDMS was increased from 4 to 12%, γ_s showed a negligible increase. The maintained low surface energy was attributable to the abundance of Si on each coating surface. It has been reported that when the surface energy is below 30 mJ m^{-2} , barnacles are difficult to adhere.⁶² Therefore, the coatings could effectively prevent the attachment of marine barnacles. These results indicated that the time-dependent behavior is related to the rearrangement of surface chemical groups when immersed in water.

Bacterial Adhesion Tests. Two representative bacteria, *Escherichia coli* and *Staphylococcus aureus*, were used to investigate antifouling property of the coatings. Figure 5a,c presents that a large amount of *E. coli* colonies adhered on the pristine PDMS coating ($42,800 \text{ cfu/cm}^2$). In contrast, there were significantly fewer colonies adhered on PPF1 (2400–

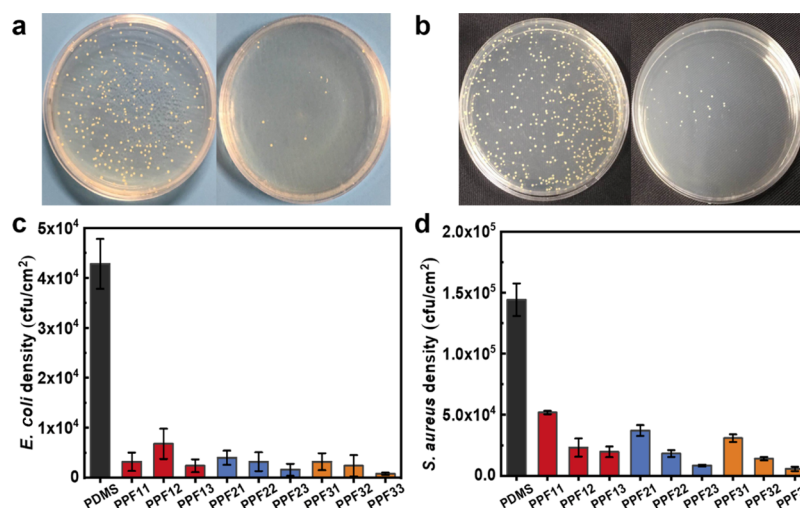


Figure 5. Bacterial adhesion tests. Images of (a) *E. coli* and (b) *S. aureus* bacterial colonies on the PDMS coating (left) and PPF33 coating (right) after 24 h exposure. (c) *E. coli* and (d) *S. aureus* numbers attached on the PDMS and PPF coating surfaces.

6800 cfu/cm²), PPF2 (1600–4000 cfu/cm²), and PPF3 (800–3200 cfu/cm²) coatings. Particularly, there were only 800 cfu/cm² colonies on PPF33, indicating a 98.1% reduction of bacterial adhesion. The resistance of *S. aureus* is also presented in Figure 5b,d. After exposure for 24 h, the *S. aureus* adhesion on PVP–PFA–PDMS coatings could still be prevented (~96.1% of reduction). These results demonstrated that the amphiphilic coatings exhibited the exceptional antibacterial properties, and the antibacterial efficiency was proportional to the PVP content.

Settlement Cells of Marine Diatoms. We used *N. parva* diatoms to test their settlement on amphiphilic surfaces. As shown in Figure 6, all PPF coating surfaces presented a

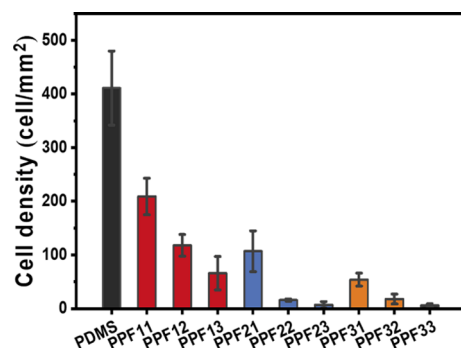


Figure 6. *N. parva* diatom adhesion. Numbers of diatoms attached to coating surfaces after 24 h ($n = 3$).

significantly lower settlement number of diatoms than the control group (PDMS). Thereinto, the PPF33 coating presented a 98.5% reduction of the diatom adhesion compared with the PDMS coating. Moreover, the efficiency of diatom resistance was proportional to the PVP content, in accordance with the previous reports stating the stronger adhesion strength of diatoms on hydrophobic rather than hydrophilic surfaces.^{60,63} The results highlighted the importance of PVP-based copolymers that resulted in the hydrophilic surfaces. Overall, it indicated that the surfaces of amphiphilic copolymer-based coatings could restructure underwater and exhibit superior fouling-release property, consistent with various previous reports.^{64,65}

Pseudobarnacle Adhesion. According to ASTM D 5618-20, we used PosiTest obtained from DeFelsko for the simulation of barnacles. In Figure 7, all of the PPF coating

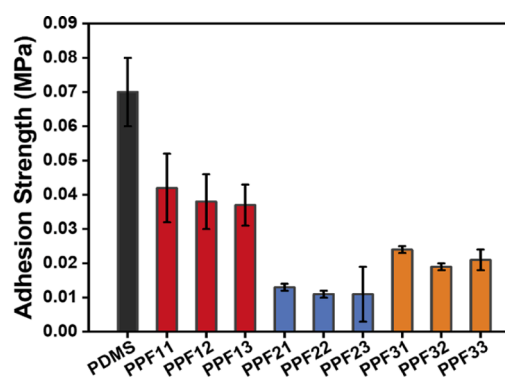


Figure 7. Pseudobarnacle adhesion strength on different coating surfaces.

samples showed a lower adhesion strength compared with the PDMS sample. There was a lowest adhesion force on the surface of PPF2 coatings, which exhibited 81.4–84.3% reduction compared with PDMS. When the molecular weight was lower (PPF1), the hydrophobic PFA segments were less on the coating surfaces and thus reducing the fouling-release efficiency. In comparison, the higher molecular weight copolymer-based coating (PPF3) surface would enrich a large amount of hydrophilic PVP segments after reconstruction, inducing a more hydrophilic surface and increasing the adhesion strength between biofouls and the surface.⁵³ So, there was an optimum Mn of PVP for the fouling-release property of the coatings. Furthermore, it showed a negligible reduction of adhesion strength on different copolymer content-based coating surfaces. It was illustrated that PPF coating could effectively decrease the adhesion strength of marine barnacles, which could be easily removed at a low ship speed.

CONCLUSIONS

In this work, we developed a set of amphiphilic marine coatings based on a novel PVP–PFA–PDMS block copolymer and systematically studied their antifouling and fouling-release

efficiency on the basis of different molecular weights and copolymer contents. In addition, the coating surface could undergo reconstruction and the PVP moieties could immigrate to the surface after 3 days of immersion in water. The amphiphilic coatings presented an outstanding antifouling property (98.1 and 98.5% reduction of bacterial and diatom adhesion, respectively) and fouling-release capability (84.3% reduction of mimic barnacle adhesion strength) compared to the pristine PDMS, and the coatings with higher percentage of PVP showed better antifouling performance. This work holds potential to produce a series of nontoxic antifouling coatings which can be applied in various marine and industrial environments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.0c02329>.

Detailed procedure of pseudobarnacle adhesion tests and synthesis process of FA and ^{19}F NMR spectra of FA and PVP–PFA–PDMS (PDF)

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[†]H.G. and P.C. contributed equally to this work. H.G., P.C., J.Y., and L.Z. conceived the study. H.G., P.C., S.T., Y.M., Q.L., and C.W. performed experiments. The manuscript was written by H.G., P.C., J.Y., and L.Z. All authors analyzed the data, discussed the results, agreed on their implications, and contributed to the preparation of the manuscript. All authors have given approval to the final version of the manuscript.

Notes

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